

Software Requirements Specification (SRS)

Standard IEEE 830-1998 Conforming Specification for AI Appliance Leasing Platform

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

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Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) establishes the foundational technical, functional, and operational requirements for the **AI-Based Household Appliances Rental Platform (RentAI / SDC2)**. The platform provides a full-stack, enterprise-grade durable goods rental marketplace characterized by pure NoSQL data persistence, machine-learning-driven customer churn mitigation, personalized collaborative filtering recommendations, and local privacy-preserving generative AI customer assistance.

1.2 Scope

The system encompasses an end-to-end multi-sided platform linking appliance owners/providers with prospective short- and medium-term tenants. It addresses the friction of substantial upfront capital expenditure for household electronics and furniture through dynamic subscription leasing, supported by automated KYC verification, digital rental contract agreements, return logistics dispatch, and predictive maintenance tracking.

1.3 Operating Environment & Stack Specifications

Layer	Technology	Role / Specification
Backend Framework	Django 6.0.7 / Django REST Framework 3.17	Microservice-ready REST API gateway
Database Engine	MongoDB 8.3 Community Edition (NoSQL)	Native Document Store via MongoEngine ODM (Zero SQL)
Machine Learning	LightGBM 4.3, Scikit-Learn 1.6, NumPy, SciPy	Churn prediction, matrix factorization FunkSVD, ARIMA
Generative AI	Ollama Local Inference Daemon (11434)	qwen2.5:0.5b / rentai-llm:latest RAG integration
Frontend Client	React 19, Vite 8.2, Tailwind CSS, Framer Motion	Single-Page Application (SPA) with responsive design
Runtime OS	Linux (Ubuntu 22.04/24.04), Windows 10/11 Pro	Decoupled asynchronous daemon processes

2. Overall Description & User Classes

2.1 User Personas

- **Tenant / Lessee:** Urban migrant, student, or corporate professional seeking flexible appliance subscriptions (1 to 24 months) without maintenance or resale burdens.
- **Appliance Owner / Vendor:** Hardware distributor or certified host listing durable appliances (refrigerators, washers, ACs, TVs) for recurring monthly yield.
- **Logistics & Field Engineer:** Technicians managing scheduled delivery, professional installation, safety grounding audits, and asset recovery.
- **System Administrator & BI Analyst:** Platform administrators monitoring financial volume, churn mitigation alerts, catalog distribution, and demand forecasts.

3. Specific Functional Requirements

3.1 Appliance Catalog & Multi-Attribute Search (FR-CAT)

The platform must index at least 1,000+ distinct appliance units spanning over 50 categories. Query filters must support simultaneous multi-criteria evaluation: categorical hierarchy, daily pricing ceilings, refundable security deposit limits, brand affinity, and geographic availability.

3.2 Machine Learning Churn Mitigation Engine (FR-ML-CHURN)

The backend shall run an automated behavioral feature extraction pipeline generating 15 normalized metrics (inactivity ratio, payment delinquency, ticket dissatisfaction, usage frequency). The system must compute real-time churn probability scores using a pre-trained LightGBM model and surface high-risk tenants (probability ≥ 0.65) to administrators with tailored retention incentives.

3.3 Tri-Paradigm Collaborative Filtering (FR-ML-REC)

The recommendation engine shall implement three distinct algorithmic paradigms: User-Based CF (cosine similarity over rental vectors), Item-Based CF (cross-appliance co-occurrence matrix), and Model-Based FunkSVD (latent matrix decomposition with bias terms). The API must return top-K personalized recommendations with a fallback to global popularity rankings for cold-start users.

3.4 Local Privacy-Preserving GenAI RAG Assistant (FR-AI-CHAT)

The system shall provide an embedded AI conversational assistant. The assistant must query local MongoDB inventory via keyword-based Retrieval-Augmented Generation (RAG) and generate

contextually grounded answers using local Ollama models (CPU optimized, zero external cloud API token expense, zero data leakage).

4. Non-Functional Requirements (NFR)

Category	Requirement	Metric / Acceptance Criteria
Performance	API Response Latency	≤ 45 ms for 95% of catalog queries; ≤ 15 ms for ML churn scoring.
Availability	System Uptime	99.9% uptime during operational hours; self-healing daemon failover.
Security	Authentication & Authorization	JWT with HMAC-SHA256, bcrypt password hashing, RBAC middleware.
Data Integrity	NoSQL ACID Leases	State machine integrity across booking status transitions.